

Hands-on Simple BLE application generated from CubeMX

Prerequisites

What we'll use

- [STM32CubeMX](#) V6.2.1 or later
- [CubeIDE](#) V1.6.0 or later
- [STM32CubeProgrammer](#) V2.7.0 or later
- [Android](#) or [iOS](#) phone with ST BLE Toolbox app V1.0.0 or later
- 1x micro USB cable
- STM32WB Nucleo, STM32WB55RG rev Y (Cut2.1). To check revision read @0xE004 2000 (see Reference manual, DBGMCU_IDCODE) via CubeProgrammer
- Install latest FUS and BLE Full Stack. From [STM32CubeWB](#) V1.11.0 or later. Installation instruction can be found in [STM32Cube_FW_WB_V1.11.0\Projects\STM32WB_Copro_Wireless_Binaries\STM32WB5x\Release_Notes.html](#)
- Tera Term (or similar terminal app)
- **LINK TO GITHUB!!!**

Bits 31:16 **REV_ID[15:0]**: Revision
0x2001 = Revision 2.1

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:0 **DEV_ID[11:0]**: Device ID
0x495: STM32WB55xx/35xx

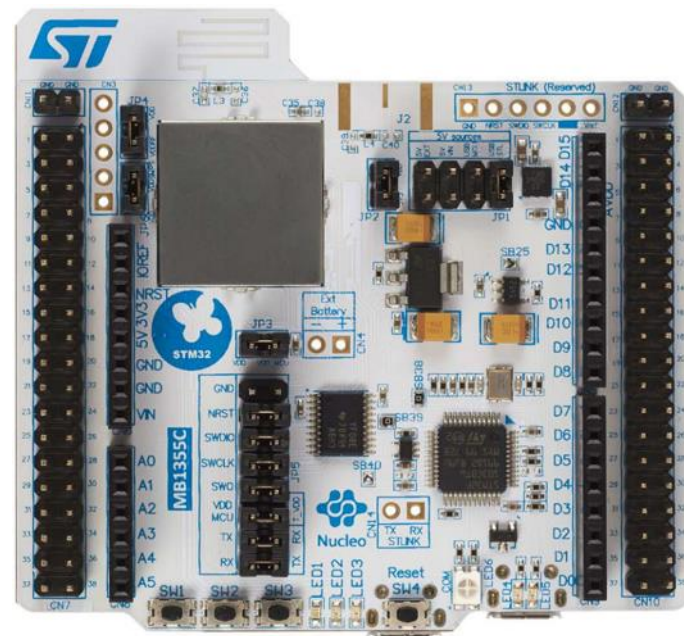
Goal



ST BLE Toolbox



GATT Client



GATT Server

- Create custom BLE Service with single characteristic with Read property
- This project will be the base for further BLE security hands-ons
- Final project can be found on Github
- Project is generated in CubeMX
- Application traces are outputted on USART1, which is connected to ST-LINK which act as USB virtual com port
- Single USB cable to ST-LINK can be used to flash/debug/display traces

Goal in more details



Peripheral

GATT Server

Custom SERVICE

Custom CHARACTERISTIC

R

Creating BLE template project in CubeMX

- Open CubeMX
- Start new project for Nucleo WB55RG (Initialize all peripherals in default mode? YES)
- System Core -> RCC: Enable HSE and LSE – Crystal/Ceramic Resonator
- System Core -> HSEM - Check Activated
- System Core -> IPCC – Check activated and enable both interrupts
- Timers -> RTC – Activate Clock, Enable internal WakeUp, enable interrupt
- Connectivity -> RF – Activate RF1
- Connectivity -> USART1 – Baud rate = 115200. Word Length 8 Bits, No parity. Connected to ST-LINK which acts as USB_VCP. Used for application traces.
 - Enable USART interrupt and add DMA for USART_TX
- Middleware -> STM32_WPAN – Check BLE
 - BLE Application and Services
 - Disable Custom P2P Server
 - Enable Custom Template
 - Configuration
 - Enable CFG_HW_USART1_ENABLED (follow the CubeMX instructions in RED)
 - Enable CFG_HW_USART1_DMA_TX_SUPPORTED
 - Set CFG_DEBUG_TRACE_UART: hw_uart1
 - Enable CFG_DEBUG_APP_TRACE

Creating BLE template project in CubeMX

- Middleware -> STM32_WPAN
 - BLE Advertising
 - Include AD_TYPE_COMPLETE_LOCAL_NAME Yes
 - BLE GATT
 - Number of services = 1
 - Service long name = "My_SVC"
 - Service short name = "My_SVC"
 - My_SVC
 - Characteristic long name = "My char"
 - Characteristic short name = "My char"
 - CHAR_PROP_BROADCAST NO
 - CHAR_PROP_READ YES
 - GATT_NOTIFY_READ_REQ_AND_WAIT_FOR_APPL_RESP No
- Clock configuration
 - Set RTC clock to LSE
 - Set RFWKP clock to LSE
- Project Manager: Generate code for CubeIDE or any other supported IDE

Code modification

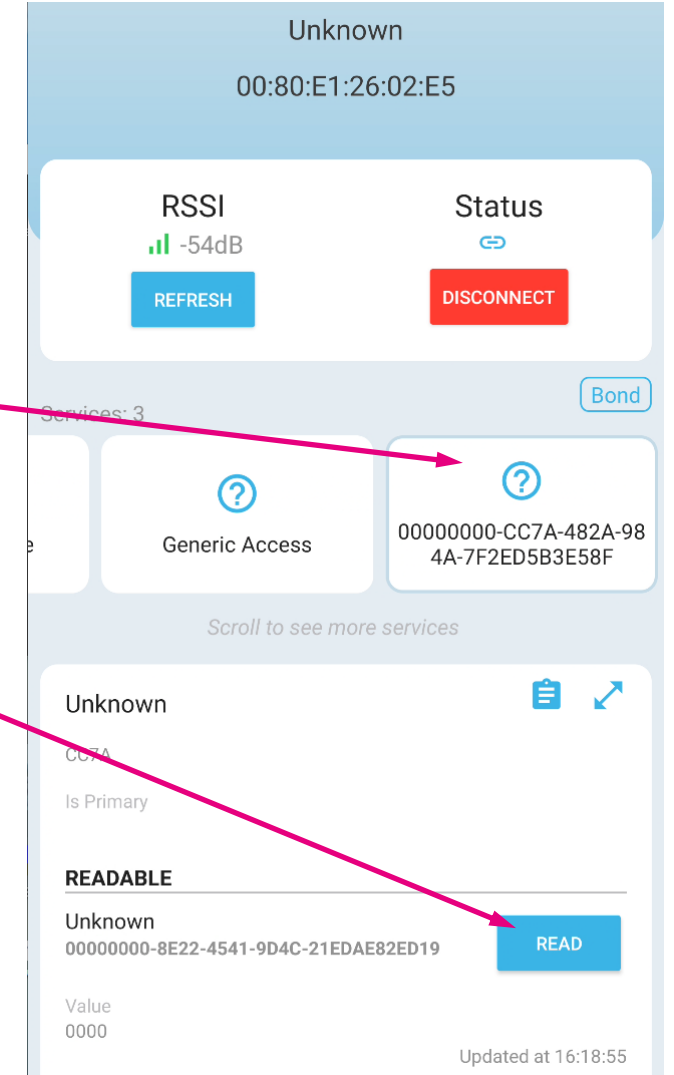
- *main.c* line 25, add `#include "stm32_seq.h"`
- *main.c* line 121, add `UTIL_SEQ_Run( UTIL_SEQ_DEFAULT);`
- *app_entry.c* line 84 `APPD_Init();`
- Compile and run
- Open Terminal
- Open ST BLE Toolbox on mobile phone
- Connect and read the Custom characteristics

Application traces

```
COM57 - Tera Term VT
File Edit Setup Control Window Help
Successfully Start Fast Advertising
HCI_LE_CONNECTION_COMPLETE_SUBEUT_CODE for connection handle 0x801
** CONNECTION UPDATE EVENT WITH CLIENT
```

Custom BLE Service

Custom BLE
Characteristic



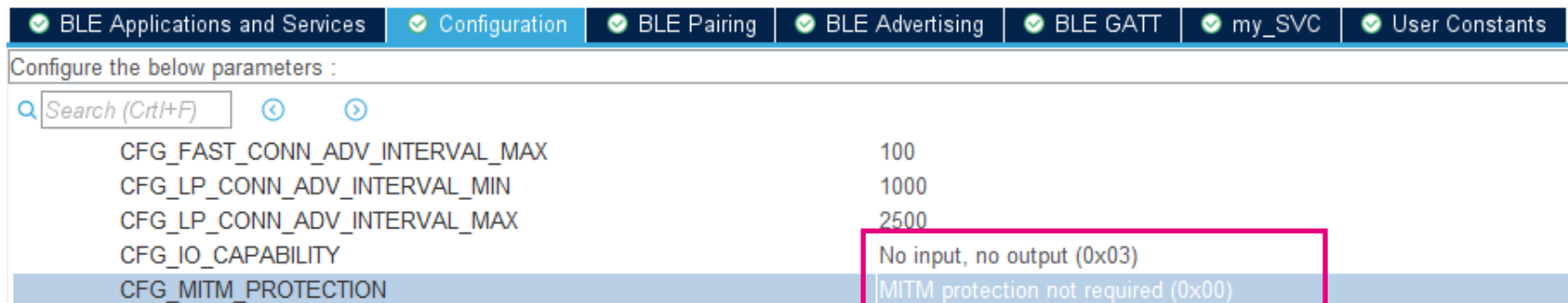
Thank you

Hands-on Pairing Just works

Pairing method selection

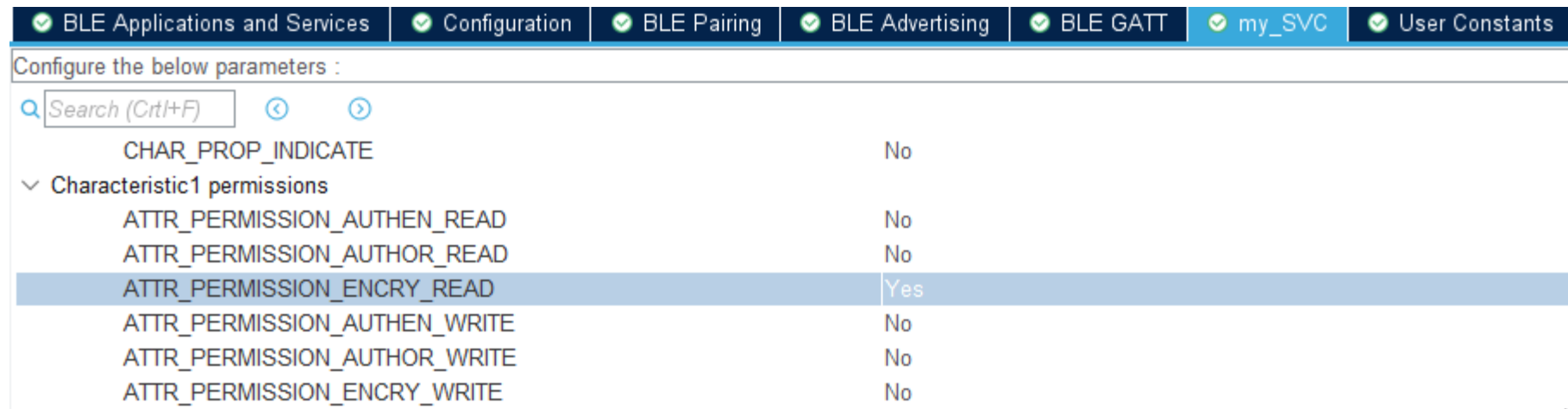
- Mobile phone indicates that it supports Keyboard and display as its IO capabilities
- To select Just works pairing, STM32WB must indicate No input/No Output and MITM protection **not** required

Initiator/ Responder	Display only	Display and Yes/No input	Keyboard only	No input/output	Keyboard and display
Display only	Just works	Just works	Passkey Entry	Just works	Passkey Entry
Display and Yes/No input	-	Just works (Legacy)	Passkey Entry	Just works	Passkey Entry (Legacy)
		Numeric comparison (SSP)			Numeric comparison (SSP)
Keyboard only	-	-	Passkey Entry	Just works	Passkey Entry
No input/output	-	-	-	Just works	Just works
Keyboard and display	-	-	-	-	Passkey Entry (Legacy)
					Numeric comparison (SSP)



Setting Characteristic permission

- Characteristic must require encryption before it can be read by Client. Upon read request, GATT server will generate error message “Insufficient Encryption”. Client will then initiate pairing and encrypt the link.
- You shall not require Authentication since there is no MITM protection with Just Works



Testing Just works pairing

- When trying to read the characteristic Android/iOS will pop up pairing message.

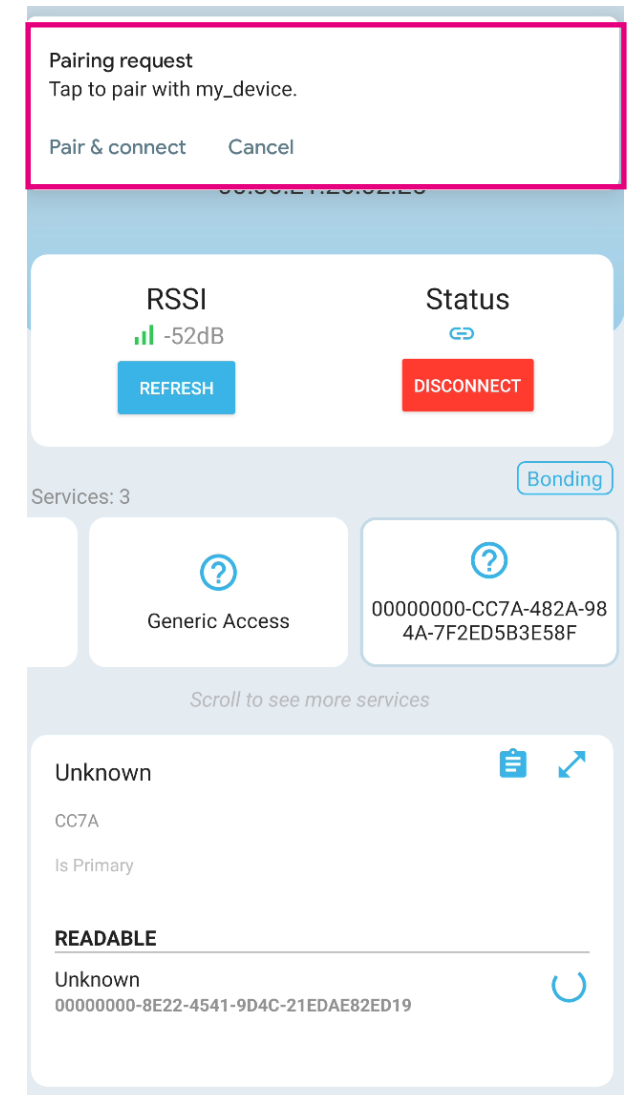
Status = 0 means no error

For more info check application note

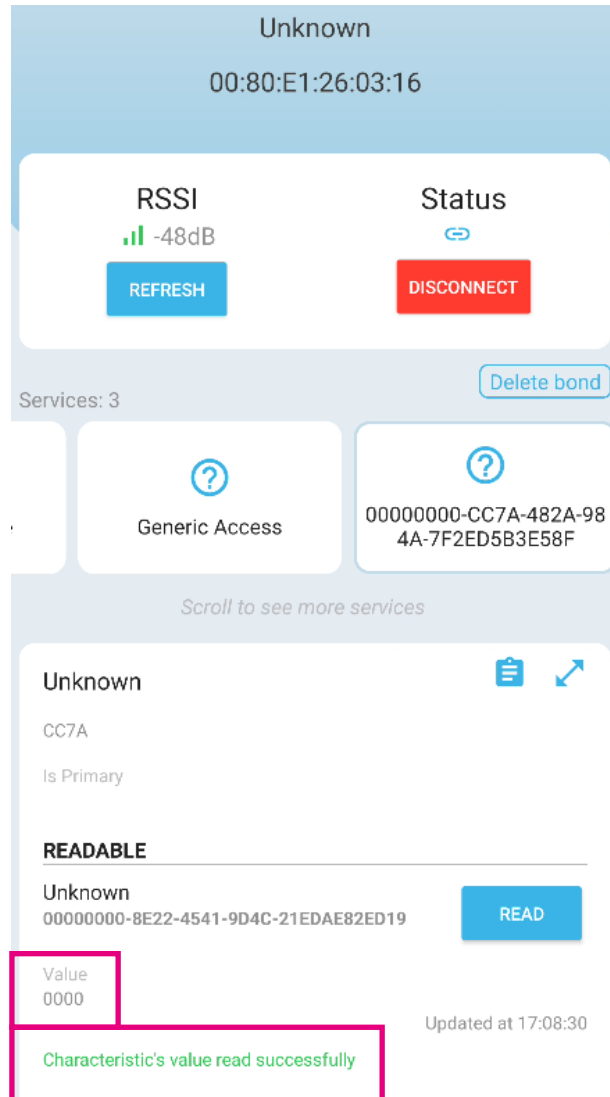
AN5270 - STM32WB Bluetooth® Low Energy (BLE) wireless interface

ACI_GAP_PAIRING_COMPLETE_EVENT

```
COM57 - Tera Term VT
File Edit Setup Control Window Help
HCI_LE_CONNECTION_COMPLETE_SUBEVENT_CODE for connection handle 0x802
*** CONNECTION UPDATE EVENT WITH CLIENT
*** CONNECTION UPDATE EVENT WITH CLIENT
BLE_CTRL_App_Notification: ACI_GAP_PAIRING_COMPLETE_USEUT_CODE, pairing_complete->Status = 0
```



Characteristic's value read successfully



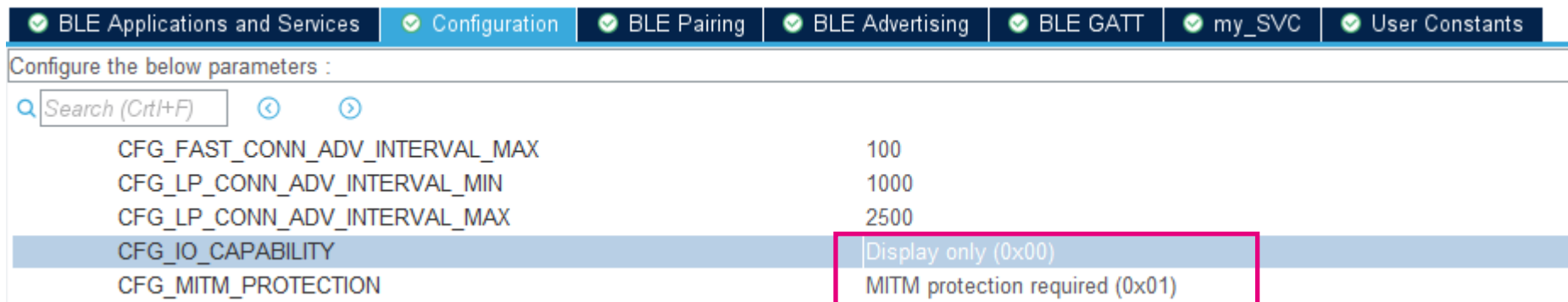
Thank you

Hands-on
Pairing
Passkey entry
Display only

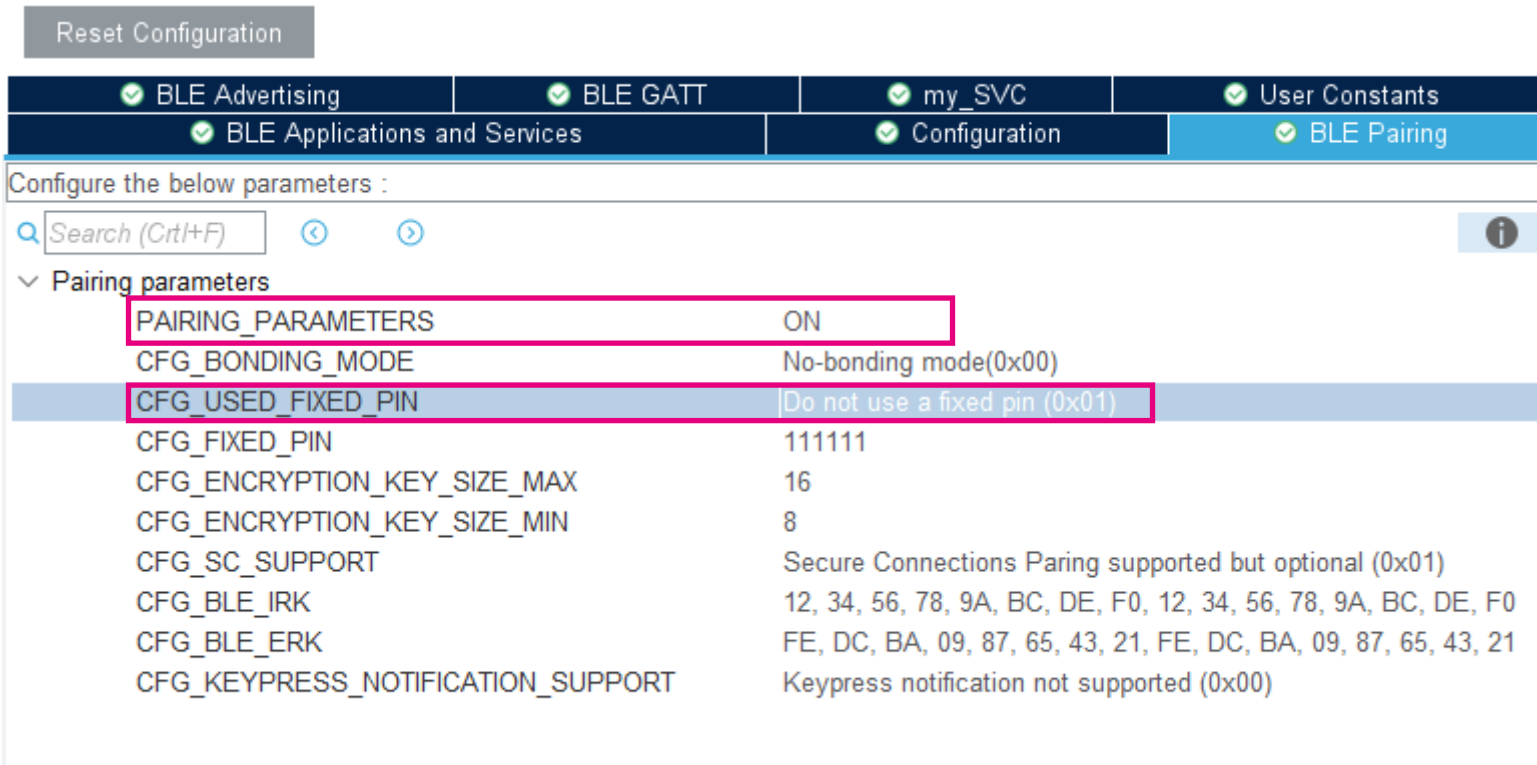
Pairing method selection

- Mobile phone indicates that it supports Keyboard and display as its IO capabilities
- STM32WB indicates support of Display only
- The 6-digit decimal number is hardcoded in the STM32WB application. We must enter it in the phone.

Initiator/ Responder	Display only	Display and Yes/No input	Keyboard only	No input/output	Keyboard and display
Display only	Just works	Just works	Passkey Entry	Just works	Passkey Entry
Display and Yes/No input	-	Just works (Legacy)	Passkey Entry	Just works	Passkey Entry (Legacy)
		Numeric comparison (SSP)			Numeric comparison (SSP)
Keyboard only	-	-	Passkey Entry	Just works	Passkey Entry
No input/output	-	-	-	Just works	Just works
Keyboard and display	-	-	-	-	Passkey Entry (Legacy)
					Numeric comparison (SSP)



Generation of 6-digit decimal number



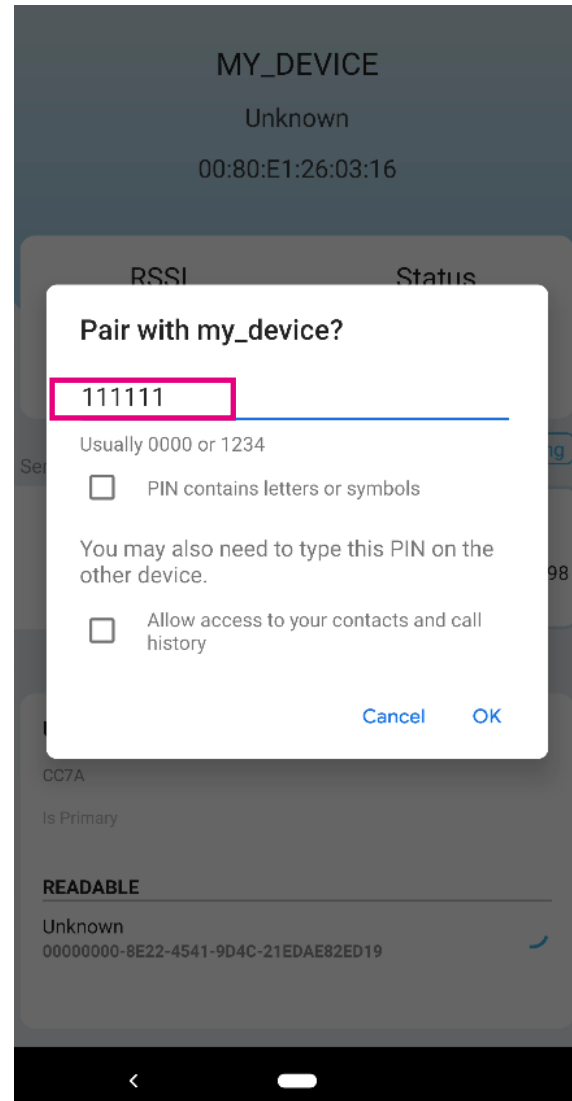
In `app_ble.c`

case `ACI_GAP_PASS_KEY_REQ_VSEVT_CODE`:

```
    aci_gap_pass_key_resp(BleApplicationContext.BleApplicationContext_legacy.connectionHandle,  
                          123456); // Line 505
```

break;

Testing Passkey entry Display only on STM32WB



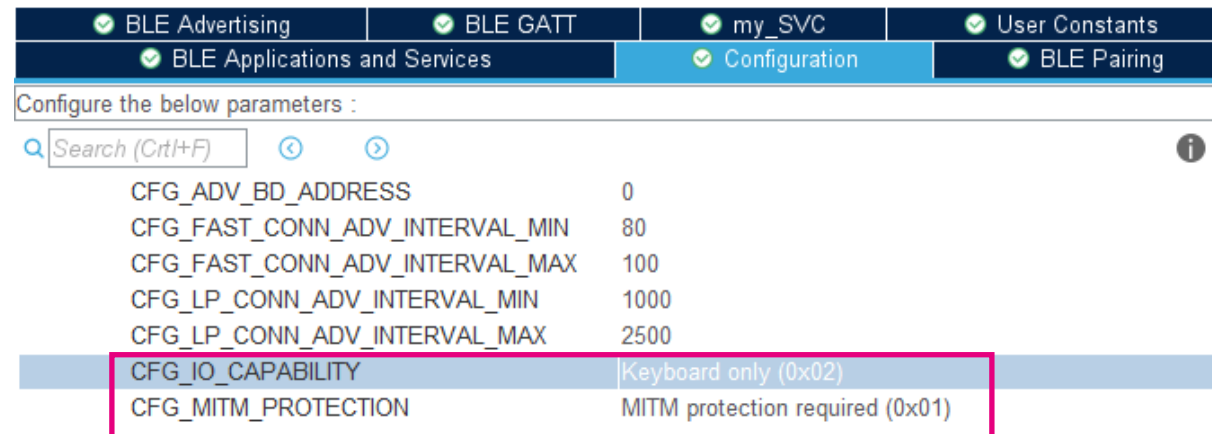
Thank you

Hands-on
Pairing
Passkey entry
Keyboard only

Pairing method selection

- Mobile phone indicates that it supports Keyboard and display as its IO capabilities
- STM32WB indicates support of Keyboard only
- Mobile phone will display 6-digit number that we must enter in STM32WB

Initiator/ Responder	Display only	Display and Yes/No input	Keyboard only	No input/output	Keyboard and display
Display only	Just works	Just works	Passkey Entry	Just works	Passkey Entry
Display and Yes/No input	-	Just works (Legacy)	Passkey Entry	Just works	Passkey Entry (Legacy)
		Numeric comparison (SSP)			Numeric comparison (SSP)
Keyboard only	-	-	Passkey Entry	Just works	Passkey Entry
No input/output	-	-	-	Just works	Just works
Keyboard and display	-	-	-	-	Passkey Entry (Legacy)
					Numeric comparison (SSP)



Testing Passkey entry Display only on STM32WB

In `app_ble.c`

```
extern UART_HandleTypeDef huart1; // Line 227
```

```
case ACI_GAP_PASS_KEY_REQ_VSEVT_CODE:; // Line 504
```

```
    uint8_t p_data[6];  
    HAL_UART_Receive(&huart1, p_data, 6, 10000);  
    p_data[0] -= 0x30; // ASCII to hex  
    p_data[1] -= 0x30;  
    p_data[2] -= 0x30;  
    p_data[3] -= 0x30;  
    p_data[4] -= 0x30;  
    p_data[5] -= 0x30;  
    uint32_t passkey = 100000*p_data[0] + 10000*p_data[1]  
    + 1000*p_data[2] + 100*p_data[3] + 10*p_data[4] + p_data[5];  
  
    aci_gap_pass_key_resp(BleApplicationContext.BleApplicationContext_legacy.connectionHandle,  
                          passkey);  
  
break;
```

COM57 - Tera Term VT

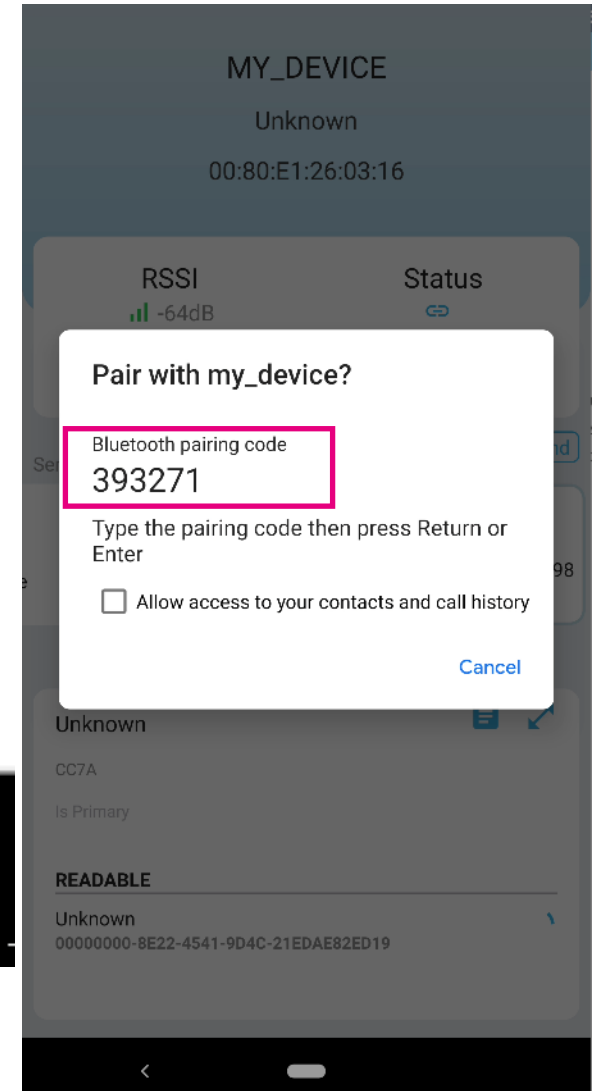
File Edit Setup Control Window Help

HCI_LE_CONNECTION_COMPLETE_SUBEVENT_CODE for connection handle 0x802

*** CONNECTION UPDATE EVENT WITH CLIENT

*** CONNECTION UPDATE EVENT WITH CLIENT

BLE_CTRL_App_Notification: ACI_GAP_PAIRING_COMPLETE_USEVT_CODE, pairing_complete->Status = 0



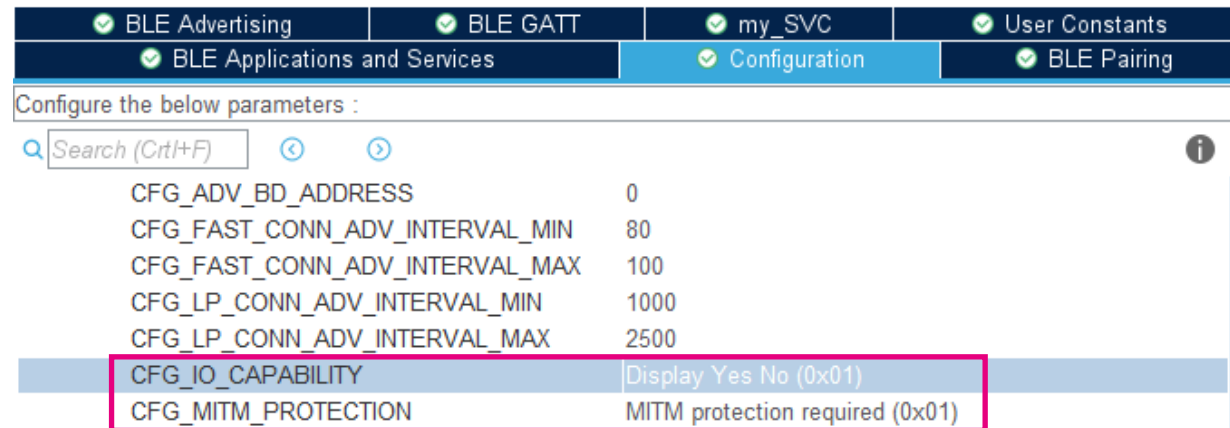
Thank you

Hands-on Pairing Numeric comparison

Pairing method selection

- Mobile phone indicates that it supports Keyboard and display as its IO capabilities
- STM32WB indicates support of Display Yes/No
- Mobile phone and STM32WB will display 6-digit number. User must confirm the number is the same on both sides.

Initiator/ Responder	Display only	Display and Yes/No input	Keyboard only	No input/output	Keyboard and display
Display only	Just works	Just works	Passkey Entry	Just works	Passkey Entry
Display and Yes/No input		Just works (Legacy)			Passkey Entry (Legacy)
	-	Numeric comparison (SSP)	Passkey Entry	Just works	Numeric comparison (SSP)
Keyboard only	-	-	Passkey Entry	Just works	Passkey Entry
No input/output	-	-	-	Just works	Just works
Keyboard and display	-	-	-	-	Passkey Entry (Legacy)
					Numeric comparison (SSP)



Testing Numeric comparison

In `app_ble.c`

```
case ACI_GAP_NUMERIC_COMPARISON_VALUE_VSEVT_CODE:// Line 508
    evt_numeric_value = (aci_gap_numeric_comparison_value_event_rp0 *)blecore_evt->data;
    numeric_value = evt_numeric_value->Numeric_Value;

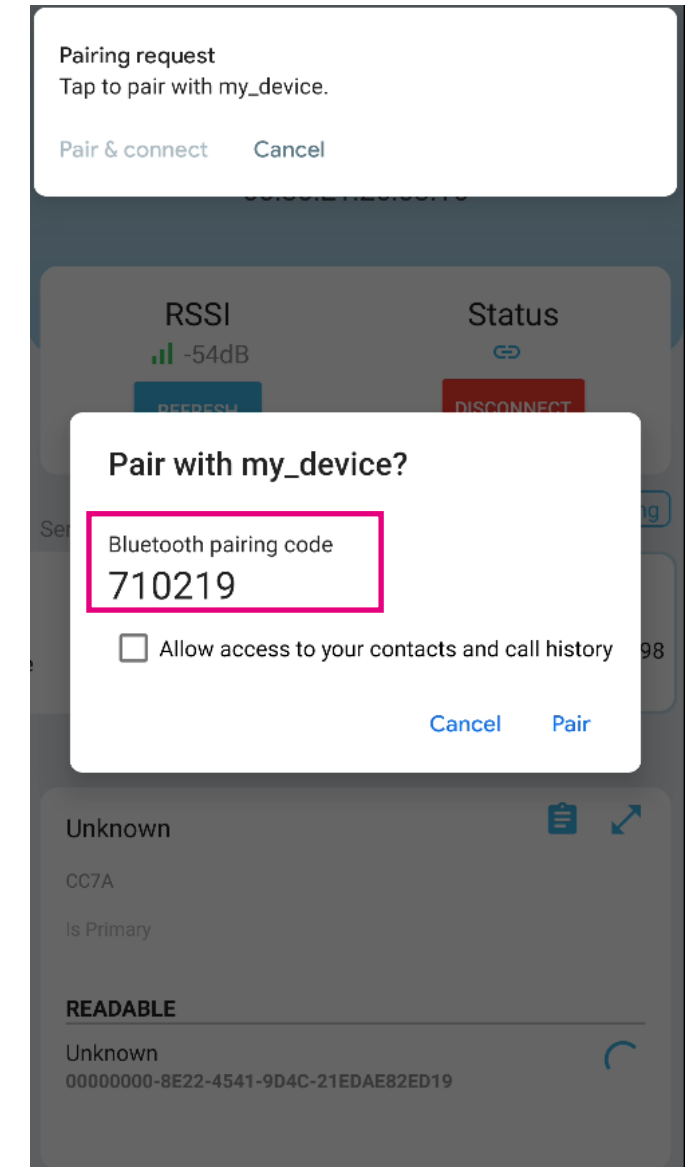
    APP_DBG_MSG("numeric_value = %d\n", numeric_value);

    aci_gap_numeric_comparison_value_confirm_yesno(BleApplicationContext.BleApplicationCon
                                                    text_legacy.connectionHandle, YES);

break;
```

```
COM63 - Tera Term VT
File Edit Setup Control Window Help
Successfully Start Fast Advertising
HCI_LE_CONNECTION_COMPLETE_SUBEVENT_CODE for connection handle 0x801

** CONNECTION UPDATE EVENT WITH CLIENT
** CONNECTION UPDATE EVENT WITH CLIENT
numeric_value = 710219
BLE_GATT_App_Notification: ACI_GAP_PAIRING_COMPLETE_USEVT_CODE, pairing_complete->Status = 0
```

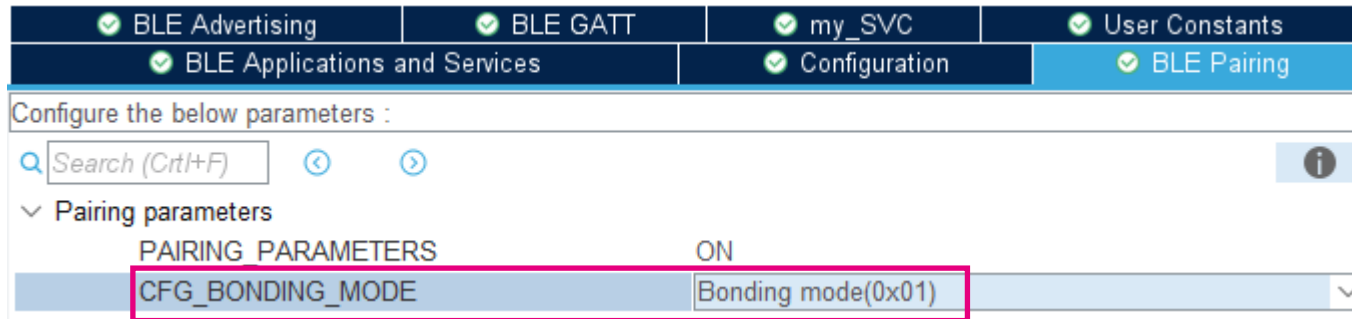


Thank you

Hands-on Bonding

Bonding

- Both Mobile phone and STM32WB must indicate support of bonding
- Pairing is done once, and the bonding information is then stored in Security database
- Subsequent connections are encrypted by keys derived from long term key



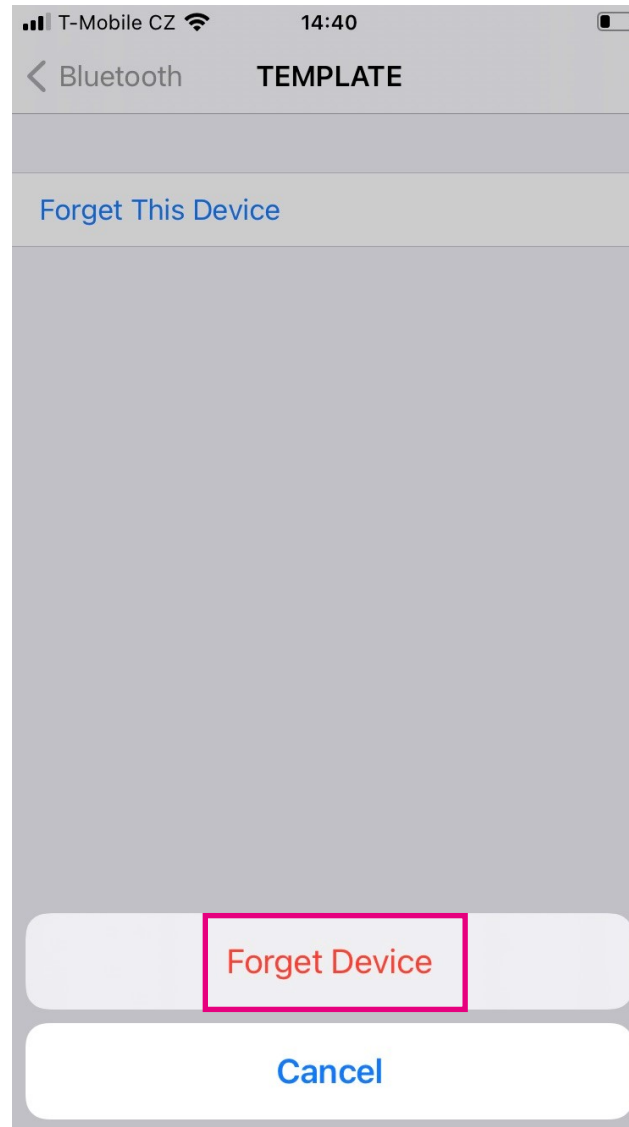
Bonding test

- Initial pairing via Numeric Comparison
- Disconnect BLE link
- Reconnect and observe automatic link encryption

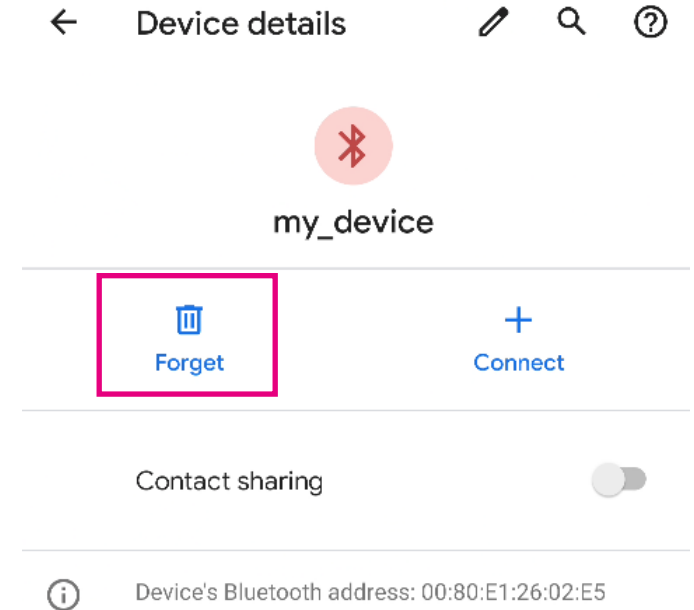
```
COM63 - Tera Term VT
File Edit Setup Control Window Help
Successfully Start Fast Advertising
HCI_LE_CONNECTION_COMPLETE_SUBEUT_CODE for connection handle 0x801
** CONNECTION UPDATE EVENT WITH CLIENT
** CONNECTION UPDATE EVENT WITH CLIENT
numeric_value = 3f21e
BLE_CTRL_App_Notification: ACI_GAP_PAIRING_COMPLETE_USEUT_CODE, pairing_complete->Status = 0
** DISCONNECTION EVENT WITH CLIENT
Successfully Start Fast Advertising
HCI_LE_CONNECTION_COMPLETE_SUBEUT_CODE for connection handle 0x802
** CONNECTION UPDATE EVENT WITH CLIENT
BLE_CTRL_App_Notification: ACI_GAP_PAIRING_COMPLETE_USEUT_CODE, pairing_complete->Status = 0
```

Erase Bonding information

- To clear the bonding information on STM32WB use
ACI_GAP_CLEAR_SECURITY_DB
- For more info check application note
AN5270 - STM32WB Bluetooth®
Low Energy (BLE) wireless interface



On iOS



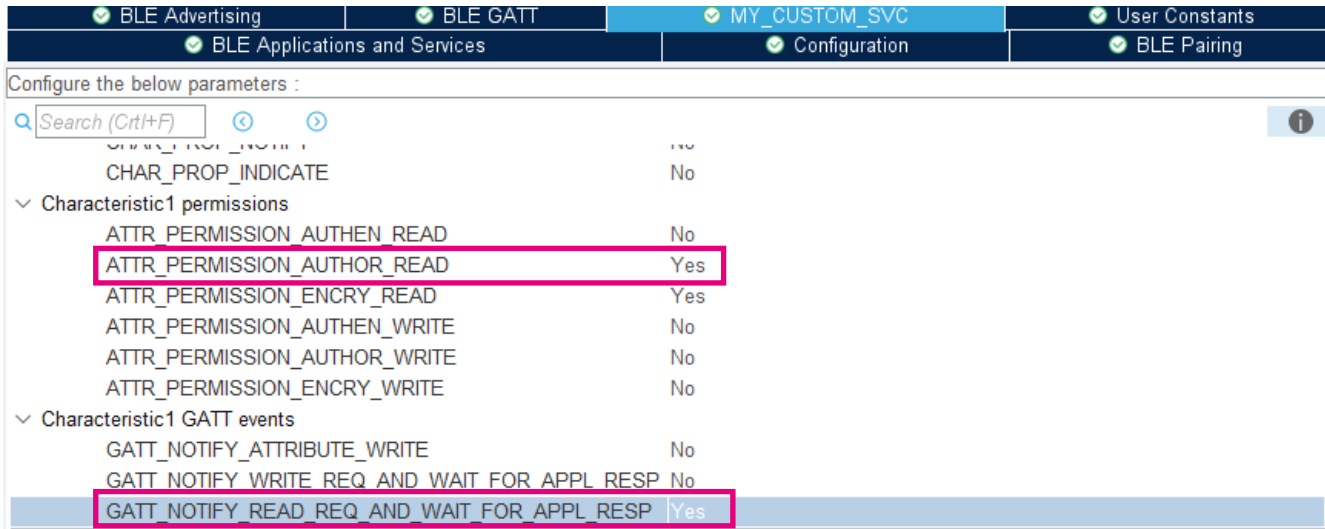
On Android

Thank you

Hands-on Authorization

Authorization

- Read or write of characteristic value by the client may require authorization by the application.
- Event generated on each Read/Write operation



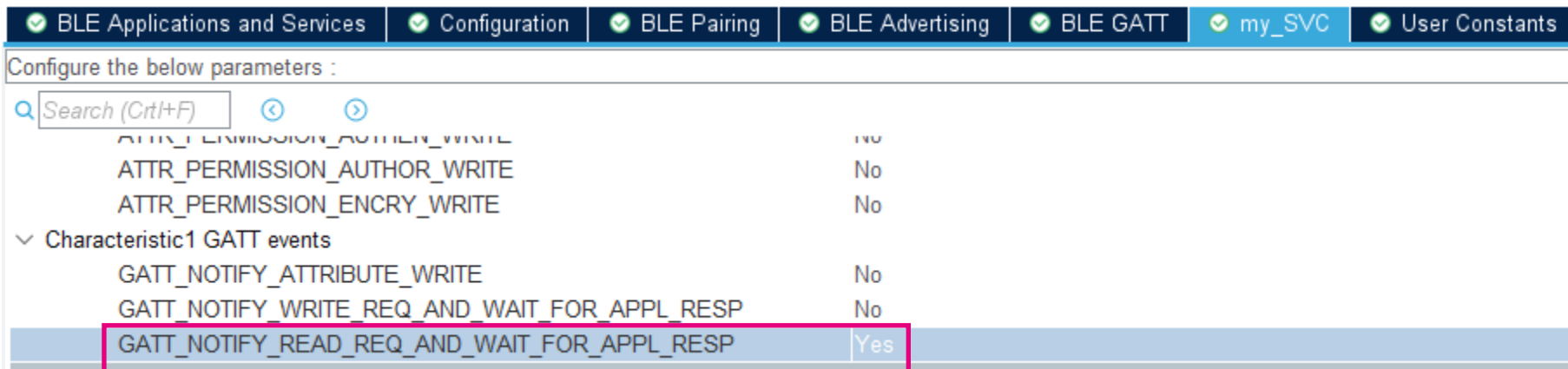
```
aci_gatt_add_char(uint16_t Service_Handle,
                  uint8_t Char_UUID_Type,
                  const Char_UUID_t* Char_UUID,
                  uint16_t Char_Value_Length,
                  uint8_t Char_Properties,
                  uint8_t Security_Permissions,
                  uint8_t GATT_Evt_Mask,
                  uint8_t Enc_Key_Size,
                  uint8_t Is_Variable,
                  uint16_t* Char_Handle )
```

Security_Permissions

- 0x02: AUTHOR_READ (Need authorization to read)
- 0x10: AUTHOR_WRITE (Need authorization to write)

Authorizing each read/write request

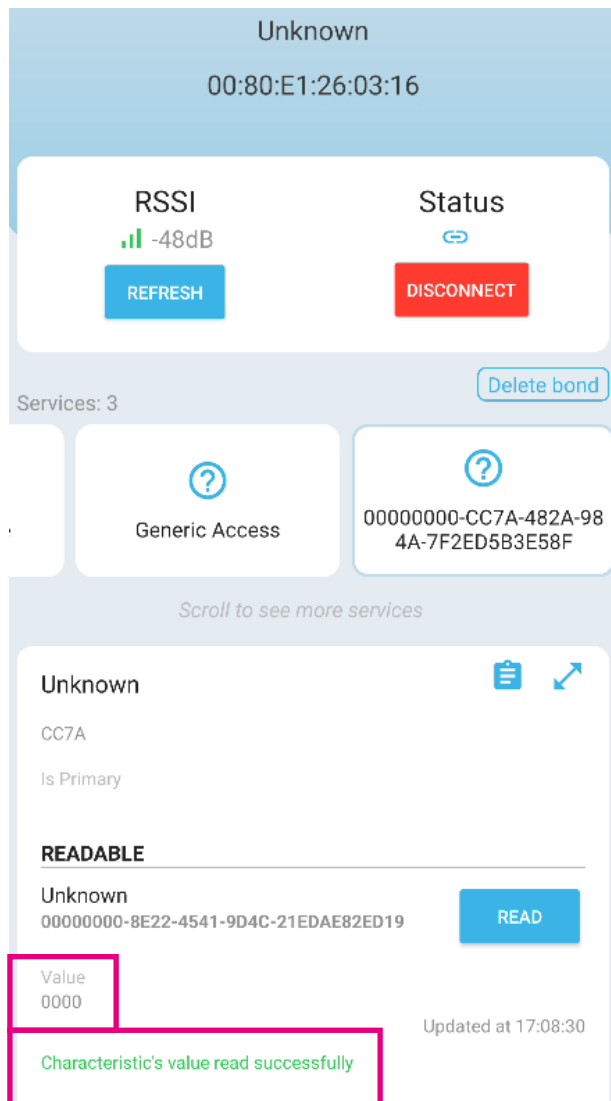
- On each Read request BLE Stack sends `ACI_GATT_READ_PERMIT_REQ_EVENT`. User application must respond with `ACI_GATT_ALLOW_READ`



In `custom_stm.c`

```
#include "custom_app.h" // Line 27
extern Custom_App_ConnHandle_Not_evt_t handleNotification; // Line 72
case ACI_GATT_READ_PERMIT_REQ_VSEVT_CODE:
    /* USER CODE BEGIN EVT_BLUE_GATT_READ_PERMIT_REQ */
    aci_gatt_allow_read( handleNotification.ConnectionHandle); // Line 143
    /* USER CODE END EVT_BLUE_GATT_READ_PERMIT_REQ */
break;
```

Characteristic's value read successfully

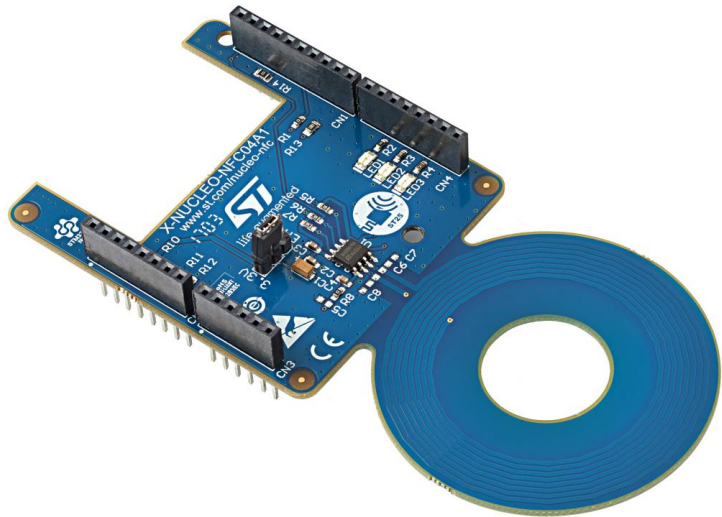


Thank you

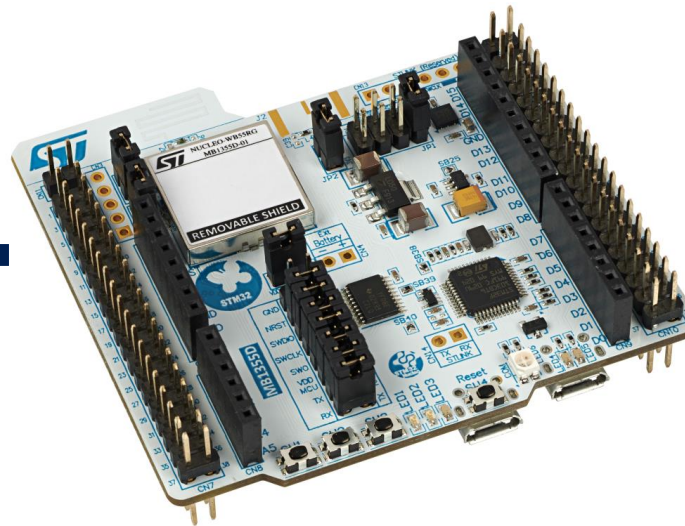
Demo Pairing OoB via NFC

Pairing OoB via NFC

- Application example of NFC pairing with STM32WB and ST25 NFC dynamic tag [link](#)
- Android application OobDemo [link](#)
- Secure Bluetooth pairing with NFC [webinar](#)



X-NUCLEO-NFC04A1



P-NUCLEO-WB55



Android app
ST25OobDemo

Thank you

Conclusion

Conclusion

- What we saw in the
 - Introduction to BLE security. Difference between Legacy and Secure Simple pairing
 - Differences between various Pairing methods
 - Examples of Security related API of BLE Stack
- What we experienced
 - Simple BLE app generated from CubeMX
 - Pairing, Just works
 - Pairing, Passkey entry Keyboard only, display only
 - Pairing Numeric comparison
 - Bonding
 - Authorization
 - Demo Pairing via NFC

Thank you

Appendix

Security related ACI command (excluding privacy)

- ACI_GATT_SET_SECURITY_PERMISSION
- ACI_GATT_SET_ACCESS_PERMISSION

Pairing

- ACI_GAP_PASS_KEY_RESP
- ACI_GAP_NUMERIC_COMPARISON_VALUE_CONFIRM_YESNO
- ACI_GAP_PASSKEY_INPUT (???)
- ACI_GAP_SLAVE_SECURITY_REQ
- ACI_GAP_SEND_PAIRING_REQ
- ACI_GAP_GET_SECURITY_LEVEL (This command can be used to get the current security settings of the device.)
- ACI_GAP_SET_IO_CAPABILITY
- ACI_GAP_SET_AUTHENTICATION_REQUIREMENT

Authorization

- ACI_GAP_SET_AUTHORIZATION_REQUIREMENT
- ACI_GAP_AUTHORIZATION_RESP
- ACI_GATT_WRITE_RESP Allow or reject a write request from a client
- ACI_GATT_ALLOW_READ / ACI_GATT_DENY_READ

OOB

- ACI_GAP_GET_OOB_DATA
- ACI_GAP_SET_OOB_DATA

Bonding

- ACI_GAP_REMOVE_BONDED_DEVICE
- ACI_GAP_CLEAR_SECURITY_DB
- ACI_GAP_ALLOW_REBOND
- ACI_GAP_IS_DEVICE_BONDED
- ACI_GAP_GET_BONDED_DEVICES

Security related ACI events (excluding privacy)

Pairing

- ACI_GAP_PAIRING_COMPLETE_EVENT
- ACI_GAP_PASS_KEY_REQ_EVENT
- ACI_GAP_AUTHORIZATION_REQ_EVENT
- ACI_GAP_SLAVE_SECURITY_INITIATED_EVENT
- ACI_GAP_BOND_LOST_EVENT
- ACI_GAP_NUMERIC_COMPARISON_VALUE_EVENT
- ACI_GAP_KEYPRESS_NOTIFICATION_EVENT

Authorization

- ACI_GATT_READ_PERMIT_REQ_EVENT
- ACI_GATT_WRITE_PERMIT_REQ_EVENT
- ACI_GATT_PREPARE_WRITE_PERMIT_REQ_EVENT

Server responds with ACI_GATT_ALLOW_READ / ACI_GATT_DENY_READ

Server responds ACI_GATT_WRITE_RESP

Server responds ACI_GATT_WRITE_RESP